

WASP-43b

R-0309

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_260316 · created 2026-07-17T22:10:16+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461115.81347 +/- 0.00072 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1825 +/- 0.0082
Transit depth (Rp/Rs)^2	3.33 +/- 0.3 [%]
Orbital Inclination (inc)	83.74 +/- 1.04
Airmass coefficient 1 (a1)	0.967 +/- 0.01
Airmass coefficient 2 (a2)	0.0206 +/- 0.0086
Scatter in the residuals of the lightcurve fit is	1.05 %
Best Comparison Star	#3 - [433, 277]
Optimal Aperture	3.25
Optimal Annulus	15.57
Transit Duration (day)	0.0567 +/- 0.0024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1825 ± 0.0082 vs 0.1594 (deviation 2.61σ)
Duration fit vs published	0.0567 d vs 0.0512 d (deviation 2.1σ)
Mid-time O–C	7.1 min
Depth SNR	20.6
Residual scatter	1.05 %

Light curve

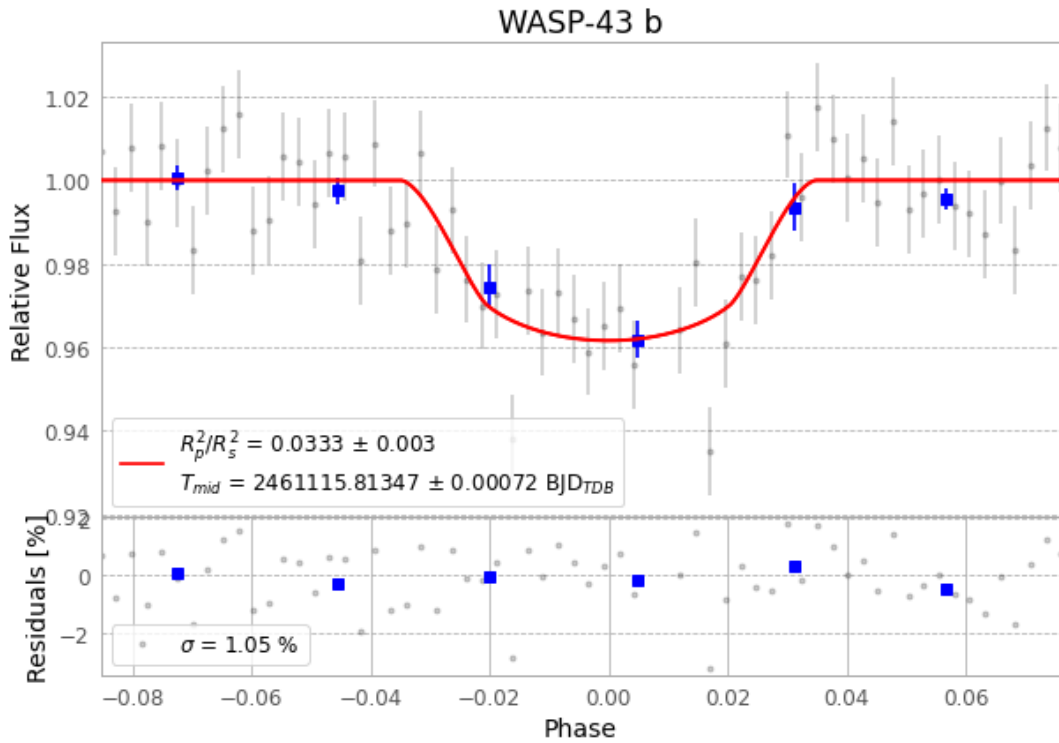


Figure 1 — FinalLightCurve_WASP-43 b_2026-03-15.png (SHA-256 00741fc40be8d477...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [412, 150], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 193978d21b532cf1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 055cf2e

Data provenance

64 FITS frames (42 MB), WASP-43260316054216.FITS → WASP-43260316085115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-260316.log (2b497f3e0a50...), FinalLightCurve_WASP-43 b_2026-03-15.png (00741fc40be8...), FinalLightCurve_WASP-43 b_2026-03-15.csv (00294bc23a83...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 22:10Z.